Waste Watch — MCU Sheet: Detailed Component Specification

Rev 0.1 — component-level wiring for the MCU reuse block. Pin assignments come from [ARCH] §3 (transcribe, don't re-derive); this document covers everything AROUND the MCU that the pin map doesn't: decoupling, clock, reset, debug, bus passives. Reference designator ranges: U1x, C1xx, R1xx, Q1xx, Y1xx, J1xx, D1xx.

U101 — STM32G0B1RET6 (LQFP64)

Supply pins & decoupling

Item	Value / part	Placement rule
Per VDD-class pin	100 nF X7R 0402, one per pin	< 2 mm from pin, via to GND at the cap pad
Bulk	4.7 µF X7R 0603, one	Near the main VDD pin
VDDA	100 nF + 1 µF X7R 0402/0603	VDDA is the ADC/analog reference — even unused, decouple it properly
VDDA feed	0 Ω 0402 from +3V3 (DNP alternative: ferrite bead BLM15 series)	Footprint lets you add a bead later if ADC ever gets used and is noisy

Wire **every** VDD/VSS pair the LQFP64 provides, plus VDDA/VSSA — and check the G0B1RE pinout for additional supply pins on this package (e.g. a separate USB/IO supply domain); if present, tie to +3V3 with its own 100 nF. This is part of the same datasheet pass that verifies PC/PD port extent and PD9.

Reset & boot

Item	Detail
NRST	100 nF to GND close to pin; net to Tag-Connect pin 3. Internal pull-up exists — no external resistor

Item	Detail
BOOT0	**No strap needed.** STM32G0 boots via nBOOT_SEL/nBOOT0 option bits — configure once in production programming. Do not copy a BOOT0 resistor from older STM32F designs

Clock

Item	Value / part	Notes
Y101	32.768 kHz crystal, **CL = 7 pF**, 3215 package (e.g. Epson FC-135R 7 pF class)	7 pF load = lowest LSE drive/power; the RTC runs 24/7
C1xx ×2	5.6–6.8 pF NP0 0402	From C = 2·(CL − C_stray), C_stray ≈ 3–4 pF. **Recalculate against the exact crystal's CL before ordering**
Layout	PC14/PC15 traces < 5 mm, GND guard ring, nothing switching adjacent	Keep LED lines and the buck's field away
System clock	Internal HSI16 (+PLL if needed) — **no HSE crystal**	One less part, no accuracy requirement justifies it; UART at 115200 from HSI is fine across temp

LSE drive level: configure LSEDRV low/medium-low in firmware for the 7 pF crystal.

Debug & programming

J101 — Tag-Connect TC2030 footprint (no connector BOM cost)

TC2030 pad	Signal	Net
1	VCC sense	+3V3
2	SWDIO	PA13
3	nRESET	NRST
4	SWCLK	PA14
5	GND	GND

TC2030 pad	Signal	Net
6	NC	— (G0 has no SWO)

Footprint: TC2030-NL (no-legs) — pads only, zero height, fits the IP67 enclosure story (programming happens before final assembly; field access is not required once OTA works). Rule: PA13/PA14 carry nothing else. They default to SWD after reset — never repurpose.

J102 — Debug UART

PA2 (USART2_TX) / PA3 (USART2_RX) + GND → 3 test pads or a 1×3 2.54 mm footprint (pads for production, header populated on EVT units). Optional 100 Ω series resistors (R1xx, DNP) if EMC testing ever fingers this stub.

I²C bus passives (bus "lives" on this sheet)

Ref	Value	Net
R110	2.2 kΩ 0402	I2C1_SCL → +3V3
R111	2.2 kΩ 0402	I2C1_SDA → +3V3

2.2 kΩ (not 4.7 k) because the bus crosses the B2B connector to the top board — extra capacitance from connector + two boards; 2.2 k keeps rise times comfortable at 400 kHz Fast-mode. Bus members: PN7160 (top board), MAX17048 (power sheet), LIS2DW12 (this sheet, optional). One set of pull-ups total — nothing on the other sheets.

LIS2DW12 accelerometer (optional-populate)

Pin	Connection
VDD, VDD_IO	+3V3, 100 nF each
SCL/SDA	I2C1 bus
INT1	PC13 (EXTI13)
INT2, unused	NC
SDO/SA0	GND (sets I²C address LSB)
CS	+3V3 (I²C mode)

Mark the whole cluster DNP-optional as a group in the BOM (chip + its two caps).

What is deliberately NOT on this sheet

- Button pull-ups (internal, configured in FW), button debounce caps (DNP pads on top board)
- LED series resistors (top board, at the LEDs)
- Level shifters, NPN stages (modem sheet)
- I²C devices other than the accel (their sheets), their decoupling (at each device)

Unused-pin policy

Any GPIO the pin map doesn't assign (spare PD/PF pins the LQFP64 turns out to have): leave unconnected in schematic, configure as analog-input in firmware init (lowest power, no floating-input current). Do NOT ground them.

ERC / capture pitfalls specific to this sheet

1. **PA11/PA12 remap trap**: the G0 family can remap PA9/PA10 onto the PA11/PA12 pins via SYSCFG. We use all four as distinct signals (PA9/10 UART, PA11 BTN8, PA12 SPARE0) — firmware must leave the remap OFF. Add a schematic text note; this is a firmware contract, not a wiring change.
2. Give the crystal nets explicit names (LSE_IN/LSE_OUT) so they're findable in layout.
3. The 37 block ports: after wiring, read the port list against capture-guide Appendix A once, aloud, before touching the root sheet.

Block BOM summary (excl. ports/labels)

Qty	Part	Refs
1	STM32G0B1RET6 LQFP64	U101
~8	100 nF 0402 X7R	C101... (per VDD pin + NRST + accel)
1	4.7 µF 0603	C110

Qty	Part	Refs
1+1	1 μF + 100 nF (VDDA)	C111, C112
1	0 Ω 0402 (VDDA feed, bead-optional)	R101
1	32.768 kHz 7 pF 3215	Y101
2	5.6–6.8 pF NP0 0402	C115, C116
2	2.2 kΩ 0402	R110, R111
1	TC2030-NL footprint	J101
1	1×3 pads/header (debug UART)	J102
1	LIS2DW12 + 2×100 nF (optional group)	U102, C130, C131